

MOLECULAR AND IONIC EQUATIONS

A chemical equation gives information about the bonds that are broken and formed or, in the case of equations involving ions, which of the ions are involved in the reaction.

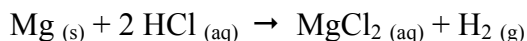
In many equations involving ions, some of the ions are not changed during the chemical reaction. They appear unchanged on both sides of the chemical equation. These ions are called *spectator ions*.

The spectator ions can be left out of the equation as they are not involved in the reaction, leaving what is known as the *net ionic equation*.

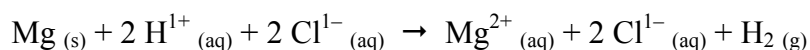
Examples

What is the net ionic equation for the reaction of hydrochloric acid with magnesium to produce magnesium chloride and hydrogen gas?

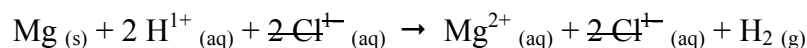
1. *Write a balanced equation in the usual way.*



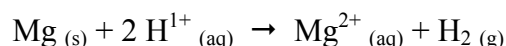
2. *Expand aqueous solutions to clearly identify ions on both sides of the equation.*



3. *Cross out ions that appear unchanged on both sides of the equation (spectator ions).*



4. *Write out the remainder of the equation - the net ionic equation.*



Lead nitrate solution, when mixed with potassium iodide solution, produces solid lead iodide and a solution of potassium nitrate.

